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CHEMISTRY | 1

1. $Na(SO_4)_2 + BaCl_2 \rightarrow$ Precipitate (PbI_2) (s)

The precipitate formed is in Yellow colour.

2. ~~NH_4SO_4~~ Ammonia gas is evolved. (NH_3).

3. Assertion is True but Reason is False.

5. Assertion is True but Reason is False.

6. In the cases of Aluminium and Zinc strips, the student would observe a displacement reaction as, these both are more reactive than $FeSO_4$ and hence will displace the Fe.

7. On heating green vitriol crystals, a reddish brown solid is left out and the smell of gas which gives off a burning Sulphur is ~~observed~~ obtained. This is Thermal Decomposition

8.
(a) Zinc Granules.
Here, Zinc displaces Hydrogen as it is more reactive and will produce Hydrogen

Calcium
Magnesium
Aluminium
Chlorine
Zinc
Iron
Tin
Lead
Hydrogen
Copper
Silver
Gold.

gas and Zinc chloride

$Zn(s) + HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$.

(b) Crushed egg shells.

In this process / reaction, dilute HCl reacts with crushed eggshells that are made of Calcium Carbonate. Hence, making the shells softer.

$HCl(aq) + CaCO_3(s) \longrightarrow$

9. The pH of a salt used to make crispy pakoras is 14. The salt type is Baking Soda i.e, Sodium ~~B~~ bicarbonate ($NaHCO_3$).

The uses of Baking Soda are:-

* It is used as an Antacid as it has ~~to~~ the properties to reduce Acidity in the Stomach.

* It also is used for cooking fried foods like Pakoras.

10. a) ~~Sodium Hydro~~ $2NaCl + 2H_2O \rightarrow 2NaOH + Cl_2 + H_2$.

The process is called Chloro-Alkali.

~~In this process Hydrogen gas is evolved at Cathode & Chlorine gas is evolved at Anode.~~

b) $Ca(OH)_2(aq) + CO_2(g) \rightarrow CaCO_3(s)\downarrow + H_2O$.

When excess Carbon Dioxide is added, it reacts with $CaCO_3$ & H_2O to form Calcium bicarbonate & since it is soluable in water, it dissolves making the lime water colourless.

11.

1. $CaO(s) + SiO_2(s) \rightarrow CaSiO_3(s)$.

	R	P
Ca	1	1
O	~~2~~3	3
Si	1	1

$\therefore$ The balanced eq^n is $CaO(s) + SiO_2(s) \rightarrow CaSiO_3(s)$.

$\therefore$ The reaction observed here is Combination reaction.

2. $TiCl_4(l) + Mg(s) \rightarrow Ti(s) + MgCl_2(s)$.

	R	P
Ti	1	1
Cl	4	$2 \times 2 = 4$
Mg	$1 \times 2 = 2$	1

$\therefore$ $TiCl_4(l) + 2Mg(s) \rightarrow Ti(s) + 2MgCl_2(s)$ is the eq^n.

$\therefore$ The reaction observed here is :- Double Displacement Reaction.

3. $H_2O_2(l) \xrightarrow{U.V} H_2O(l) + O_2(g)$.

	R	P
H	$2 \times 2 = 4$	$2 \times 2 = 4$
O	2	3

$\therefore$ $2H_2O_2(l) \xrightarrow{U.V} 2H_2O(l) + O_2(g)$.

$\therefore$ The reaction observed here is Decomposition Reaction

(4) $Pb_3O_4 + HCl \rightarrow PbCl_2 + Cl_2 + H_2O$.

	R	P
Pb	3	1
O	4	1×4
H	1×8	2×48
Cl	1	4

∴ $Pb_3O_4 + \underline{8}HCl \rightarrow \underline{3}PbCl_2 + Cl_2 + \underline{4}H_2O$.

	R	P
Pb	3	3
O	4	4
H	8	8
Cl	8	8

∴ $Pb_3O_4 + \underline{8}HCl \rightarrow \underline{3}PbCl_2 + Cl_2 + \underline{4}H_2O$.

∴ This is a Redox Reaction.

5 $KCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + KNO_3(aq)$.

K	1	1
Cl	1	1
Ag	1	1
NO_3	1	1

∴ $KCl(aq) + AgNO_3(aq) \rightarrow AgCl(s) + KNO_3(aq)$.
is the balanced equation.

∴ Displacement Reaction has occured here.